FU2 Beta - Dice Probability Analysis

Action Dice (0–10) vs Danger Dice (0–10)

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2 Introductions

These are the result from asking claude.ai about the probabilities in FU2 Beta by Nathan Russell
• Probabilities computed via exact multinomial DP

3 Probability

How to read these tables
Rows = number of Action Dice (AD). Columns = number of Danger Dice (DD). Each cell shows the probability of that outcome for a given pool. The mechanic: each DD cancels one AD showing the same face value (1–6). The highest remaining AD is the result. If all AD are cancelled, the result is a Botch.

Oracle results: 6 = Yes and… (Boon if 2+ sixes remain) • 5 = Yes… • 4 = Yes but… • 3 = No but… • 2 = No… • 1 = No and… (Botch)

3.1 Table 1 — Any Yes

Probability of result ≥ 4 (Yes but… / Yes… / Yes and…)

AD \ DD	0	1	2	3	4	5	6	7	8	9	10
0	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
1	50.0%	41.7%	34.7%	28.9%	24.1%	20.1%	16.7%	13.9%	11.6%	9.7%	8.1%
2	75.0%	66.7%	58.6%	51.0%	44.1%	38.0%	32.6%	27.9%	23.8%	20.2%	17.2%
3	87.5%	81.3%	74.3%	67.1%	60.0%	53.2%	46.8%	41.0%	35.8%	31.1%	26.9%
4	93.8%	89.6%	84.4%	78.4%	72.0%	65.5%	59.1%	52.9%	47.1%	41.7%	36.7%
5	96.9%	94.3%	90.7%	86.1%	80.9%	75.2%	69.2%	63.2%	57.3%	51.6%	46.3%
6	98.4%	96.9%	94.5%	91.2%	87.2%	82.5%	77.3%	71.9%	66.3%	60.7%	55.2%
7	99.2%	98.3%	96.8%	94.5%	91.6%	87.9%	83.6%	78.9%	73.8%	68.6%	63.3%
8	99.6%	99.1%	98.2%	96.7%	94.5%	91.8%	88.4%	84.5%	80.1%	75.4%	70.5%
9	99.8%	99.5%	98.9%	98.0%	96.5%	94.5%	91.9%	88.8%	85.1%	81.0%	76.7%
10	99.9%	99.7%	99.4%	98.8%	97.8%	96.4%	94.5%	92.0%	89.0%	85.6%	81.8%

3.2 Table 2 — Any No

Probability of result ≤ 3 or Botch (No but… / No… / No and…)

AD \ DD	0	1	2	3	4	5	6	7	8	9	10
0	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
1	50.0%	58.3%	65.3%	71.1%	75.9%	79.9%	83.3%	86.0%	88.4%	90.3%	91.9%
2	25.0%	33.3%	41.4%	49.0%	55.9%	62.0%	67.4%	72.1%	76.2%	79.8%	82.8%
3	12.5%	18.8%	25.7%	32.9%	40.0%	46.8%	53.2%	59.0%	64.2%	68.9%	73.1%
4	6.3%	10.4%	15.6%	21.6%	28.0%	34.5%	40.9%	47.1%	52.9%	58.3%	63.3%
5	3.1%	5.7%	9.3%	13.9%	19.1%	24.8%	30.8%	36.8%	42.7%	48.4%	53.7%
6	1.6%	3.1%	5.5%	8.8%	12.8%	17.5%	22.7%	28.1%	33.7%	39.3%	44.8%
7	0.8%	1.7%	3.2%	5.5%	8.4%	12.1%	16.4%	21.1%	26.1%	31.4%	36.7%
8	0.4%	0.9%	1.9%	3.3%	5.5%	8.2%	11.6%	15.5%	19.9%	24.6%	29.5%
9	0.2%	0.5%	1.1%	2.0%	3.5%	5.5%	8.1%	11.2%	14.9%	18.9%	23.3%
10	0.1%	0.3%	0.6%	1.2%	2.2%	3.6%	5.5%	8.0%	10.9%	14.4%	18.2%

3.3 Table 3 — Botch

Probability all Action Dice are cancelled (worst outcome, No and…)

AD \ DD	0	1	2	3	4	5	6	7	8	9	10
0	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
1	0.0%	16.7%	30.6%	42.1%	51.8%	59.8%	66.5%	72.1%	76.7%	80.6%	83.8%
2	0.0%	0.0%	5.1%	12.8%	21.6%	30.6%	39.2%	47.2%	54.4%	60.8%	66.5%
3	0.0%	0.0%	0.0%	2.1%	6.5%	12.7%	19.8%	27.5%	35.1%	42.5%	49.4%
4	0.0%	0.0%	0.0%	0.0%	1.1%	3.8%	8.1%	13.7%	20.2%	27.1%	34.0%
5	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	2.4%	5.6%	10.0%	15.4%	21.4%
6	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.4%	1.7%	4.0%	7.6%	12.1%
7	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	1.2%	3.0%	5.9%
8	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.9%	2.4%
9	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.7%
10	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%

3.4 Table 4 — Boon

Probability of 2+ uncancelled 6s remaining (Yes and… with extra advantage)

AD \ DD	0	1	2	3	4	5	6	7	8	9	10
0	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
1	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
2	2.8%	2.3%	1.9%	1.6%	1.3%	1.1%	0.9%	0.8%	0.7%	0.5%	0.5%
3	7.4%	6.3%	5.3%	4.5%	3.8%	3.2%	2.7%	2.3%	1.9%	1.6%	1.4%
4	13.2%	11.3%	9.6%	8.2%	7.0%	6.0%	5.1%	4.3%	3.7%	3.1%	2.7%
5	19.6%	16.9%	14.6%	12.6%	10.9%	9.4%	8.1%	6.9%	6.0%	5.1%	4.4%
6	26.3%	23.0%	20.0%	17.5%	15.2%	13.2%	11.5%	10.0%	8.7%	7.5%	6.5%
7	33.0%	29.1%	25.6%	22.6%	19.8%	17.4%	15.3%	13.4%	11.7%	10.3%	9.0%
8	39.5%	35.2%	31.3%	27.8%	24.6%	21.8%	19.3%	17.1%	15.1%	13.3%	11.7%
9	45.7%	41.1%	36.8%	33.0%	29.5%	26.3%	23.5%	20.9%	18.6%	16.5%	14.7%
10	51.5%	46.7%	42.2%	38.1%	34.4%	30.9%	27.8%	24.9%	22.3%	20.0%	17.9%

4 Analysis — Key Insights

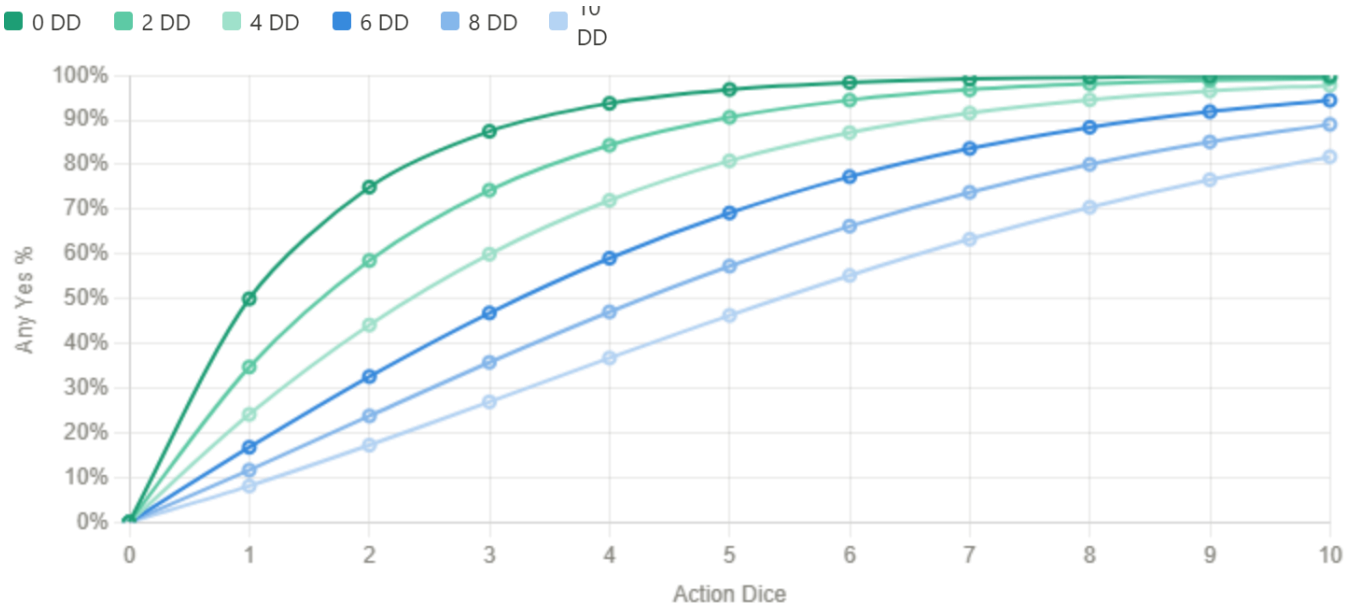
4.1 The baseline is perfectly balanced — by design

With 1 AD and 0 DD, you have exactly a 50/50 chance of Yes vs No. This is clearly intentional: the single-die roll is the game's anchor, and every modifier pulls you away from that coin flip in a meaningful, calculable way.

4.2 Extra Action Dice add more than you'd expect

Each additional AD dramatically improves odds because only the highest remaining die matters. Going from 1 to 2 AD (with no DD) jumps Yes from 50% to 75% — a 25-point gain. From 2 to 3 adds another 12.5 points to 87.5%. Returns diminish but remain significant all the way to 10 AD.

- 1 AD / 0 DD: 50.0% Yes
- 2 AD / 0 DD: 75.0% Yes (+25 points)
- 3 AD / 0 DD: 87.5% Yes (+12.5 points)
- AD / 0 DD: 96.9% Yes
- 10 AD / 0 DD: 99.9% Yes



4.3 Danger Dice are weaker than Action Dice — symmetry is an illusion

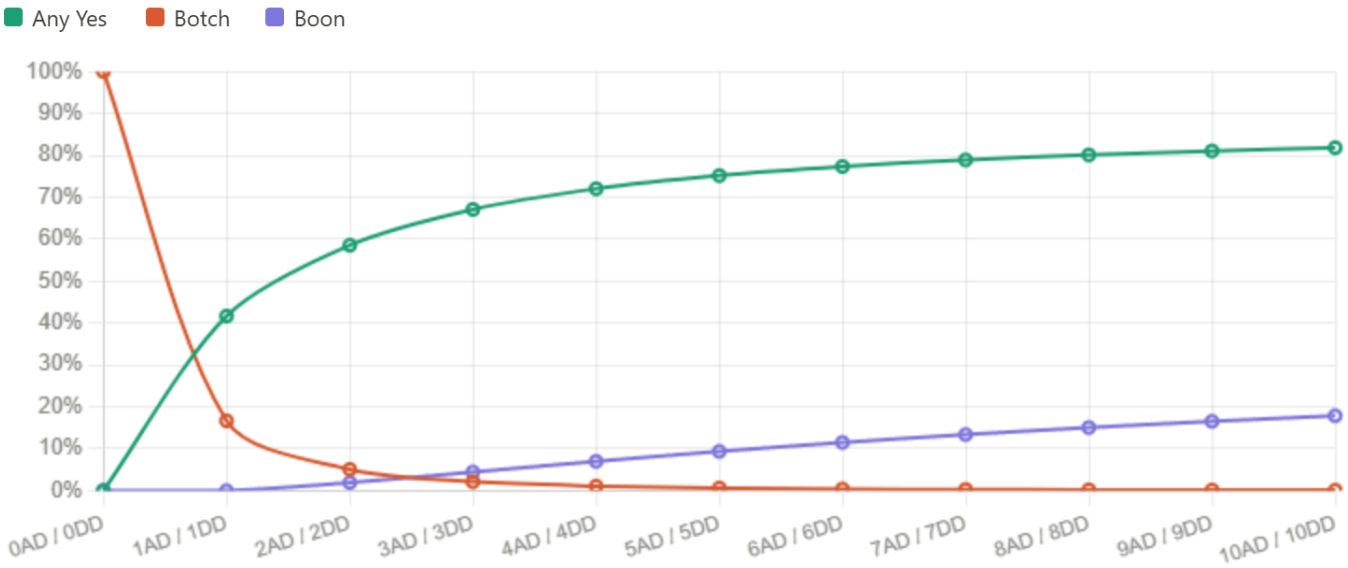
Adding a DD does not reduce your chances as much as adding an AD improves them. With 3 AD / 0 DD you have 87.5% Yes. Adding 3 DD drops it only to 67.1% — still solidly in Yes territory. DD can only cancel an AD showing the same face, so they are probabilistically inefficient at suppressing success. The asymmetry fundamentally favours players.

4.4 Equal pools strongly favour the Action player

When AD = DD, the Yes probability climbs steadily with pool size rather than hovering near 50%:

- 1 AD / 1 DD: 41.7% Yes
- 2 AD / 2 DD: 58.6% Yes
- AD / 4 DD: 72.0% Yes

- AD / 7 DD: 78.9% Yes
- 10 AD / 10 DD: 81.8% Yes

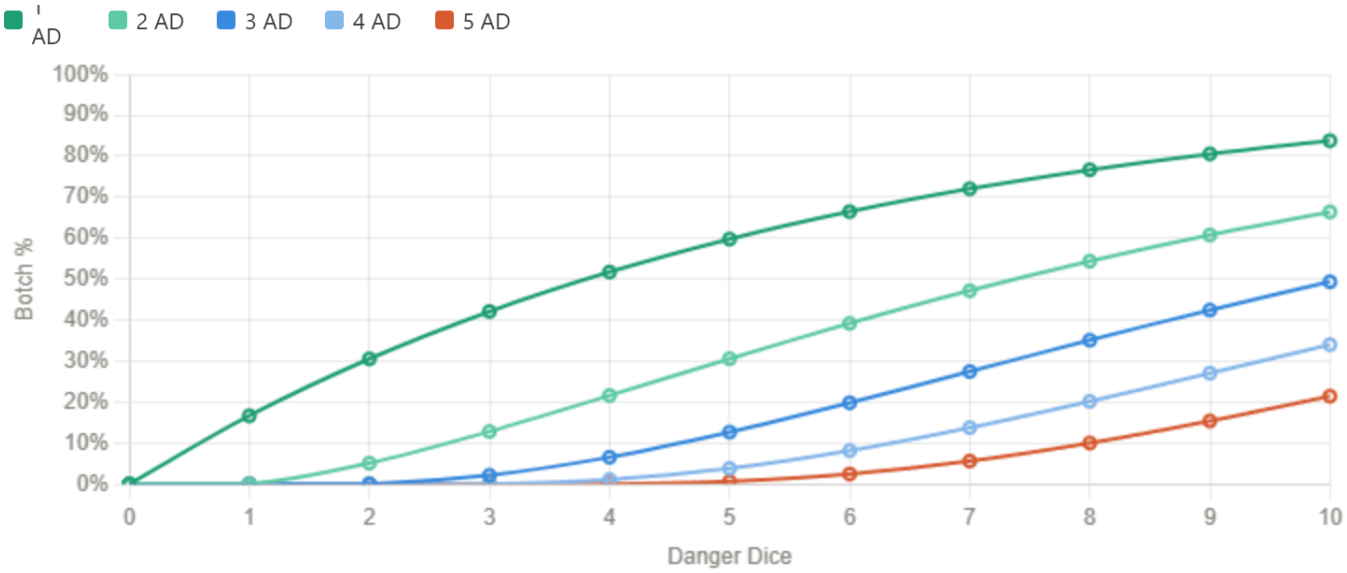


This happens because cancellation is face-specific — matching 6s cancel but matching 1s cancel too, and the high-value AD faces have a good chance of surviving even when the pools are equal. Botches on the equal diagonal fall below 2% by 4v4 and become negligible beyond that.

When AD = DD, the Yes probability is far higher than 50% and keeps rising with pool size. At 1v1 it's 41.7%, but by 5v5 it's already 75.2%, and by 10v10 it's 81.8%. This is because cancellation is face-specific — matching 6s cancel but matching 1s cancel too, so even when all dice cancel each other out, there's a good chance some high AD survive. Equal pools strongly favour the Action player. Botches (requiring *all* AD cancelled) fall below 1% by 4v4 and become vanishingly rare. Meanwhile Boons climb steadily — 10v10 produces a boon nearly 1 in 5 rolls.

4.5 Botch is a rare catastrophe, not a regular risk

Botch only becomes a real concern when DD substantially outnumber AD. With 2 AD you need around 5+ DD before Botch crosses 30%. With 4 AD, even 10 DD only produces a 34% Botch rate. The system is deliberately skewed to avoid total failures — catastrophic outcomes are reserved for situations of maximum peril where the GM has stacked multiple disadvantages.



4.6 Boons become reliable at higher Action Dice counts

Boons (2+ uncancelled 6s) require at least 2 AD but climb quickly from there. Even under moderate DD pressure, Boons become a meaningful regular occurrence:

- 2 AD / 0 DD: 2.8% Boon chance
- AD / 0 DD: 19.6% Boon chance
- 8 AD / 0 DD: 39.5% Boon chance
- 10 AD / 0 DD: 51.6% Boon chance
- 8 AD / 4 DD: 24.6% Boon chance (still significant under pressure)

This means Boons are a meaningful reward for stacking advantages — not a rare windfall. At the high end of the AD spectrum, a Boon is more likely than not.

5 Practical GM guidance: the comfortable zone

The clearest way to read these tables is through what a "fair fight" looks like. When a player has 3 AD and faces 2 DD — a common scenario of a capable character in a mildly adverse situation — their Yes rate is 74.3% and their Botch rate is 0%. That's a healthy, drama-friendly distribution: they'll usually succeed, sometimes with complications, almost never catastrophically fail. The system only becomes genuinely threatening (sub-50% Yes) when DD exceeds AD, which requires the GM to deliberately stack disadvantages.

The key design insight is that FU2 is structurally optimistic: the dice are biased toward player success, with setbacks delivered mainly through the quality of the Yes (Yes but…) rather than through outright failure. Catastrophic failure — the Botch — is effectively reserved as a dramatic device for moments of maximum peril.

The most common scenario — a capable character (3 AD) facing a mildly adverse situation (2 DD) — produces 74.3% Yes with 0% Botch. This is the system’s sweet spot: players succeed most of the time, with complications delivered through Yes but… rather than outright failure.

Yes drops below 50% only when DD exceeds AD, requiring the GM to deliberately stack multiple disadvantages. This means FU2 is structurally optimistic: setbacks arrive mainly through the quality of success, not through failure, and catastrophe is something the GM must actively engineer.

5.1 The core question a GM should ask

Not "how hard is this?" but "what is the net effect I want this situation to have on the player's odds?" That reframe is important because DD don't work like a difficulty slider — they interact with the player's AD pool in a non-linear way.

5.1.1 The three meaningful DD thresholds

Looking at the data, there are three distinct regimes based on the relationship between DD and the player's AD:

- DD < AD — this is the *comfortable zone*. The player retains a strong Yes majority in almost all cases. Even 3 DD against 4 AD leaves the player at 78.4% Yes and 0% Botch. Use this for challenges that should feel tense but rarely derail things.

- DD = AD — this is the *contested zone*. As shown earlier, equal pools still favour the player (58–82% Yes depending on pool size), but the fiction feels genuinely uncertain. Botch risk stays low. This is the sweet spot for "real" dramatic moments where failure is plausible.

- DD > AD — this is the *danger zone*. Once DD exceeds AD, Yes drops below 50% and Botch risk climbs meaningfully. With 2 AD and 5 DD the player is below 38% Yes. Reserve this for situations that should feel desperate or near-impossible.

5.1.2 A working ruleset: assign DD by source, not by feel

The most consistent approach is to treat DD as modular additions from distinct sources rather than a single holistic difficulty number. Each source adds 1–2 DD:

- *Environment* — difficult terrain, poor lighting, bad weather, time pressure. Contributes 1 DD per significant factor.

- *Opposition* — the relevant tags of the threat or NPC. A "heavily armoured" guard adds 1 DD to attack rolls. A "master swordsman" adds 1 DD to contests. Tags should add DD directly — this is actually how FU2 intends it.

- *Character condition* — injuries, exhaustion, fear, being outnumbered. Conditions on the character add 1 DD each to relevant actions.

- *Situational disadvantage* — acting without the right tools, improvising under fire, attempting something outside the character's competence. 1 DD per meaningful gap.
  This modular approach has a natural ceiling. If you list out genuine situational factors and arrive at 4+ DD for a typical player with 2–3 AD, that's the fiction telling you the situation is nearly hopeless — and that should either feel intentional (the players walked into a trap) or be a signal to narrate a way out rather than rolling.

5.1.3 The most important rule: cap at the fiction

The modular approach has a natural self-limiting property. If you honestly list out every genuine factor working against the player and end up at 5+ DD for a character with 2 AD, the fiction is telling you this action is nearly suicidal (20–30% Yes). That's either intentional — the players made a reckless choice — or it's a signal to reframe: maybe the player shouldn't be rolling at all and should instead be narrating how they get into a better position first.

5.1.4 Watch for these common GM traps

Assigning DD emotionally rather than from tags is the main pitfall. "This feels hard" is not a DD. "The enemy has the *veteran soldier* tag and the player is *wounded*" is two DD — and that's the whole picture. If you find yourself routinely assigning 3+ DD, check whether you're actually expressing the fiction or just making things harder out of habit.

The other trap is forgetting that DD don't scale linearly. Going from 2 DD to 3 DD against a 3 AD player drops them from 74% to 67% Yes — a meaningful but not dramatic shift. But going from 0 DD to 1 DD against a 1 AD player drops them from 50% to 42% while spiking their Botch chance from 0% to 17%. Low-AD players are disproportionately punished by even a single DD, which is worth remembering when less experienced or weaker characters are at the table.

5.2 Quick reference

5.2.1 Common scenarios

Key breakpoints for table reference at a glance:

Scenario	AD	DD	Any Yes	Botch	Boon
Bare minimum	1	0	50.0%	0.0%	0.0%
Slight advantage	2	0	75.0%	0.0%	2.8%
Good position	3	0	87.5%	0.0%	7.4%
Strong character	4	0	93.8%	0.0%	13.2%
Versus 1 hazard	2	1	66.7%	0.0%	2.3%
Even contest	3	3	67.1%	2.1%	4.5%
Outmatched	2	4	44.1%	21.6%	1.3%
Severely outmatched	1	4	24.1%	51.8%	0.0%
Maximum danger	2	8	23.8%	54.4%	0.7%

5.2.2 Concrete benchmarks to internalise

DANGER DICE BY COUNT — WHAT EACH LEVEL MEANS

0 DD

Uncontested

No meaningful opposition. Success is nearly certain for any trained character. Reserve for colour scenes, downtime actions, or trivially easy tasks.

1 DD

Mildly adverse

One real factor working against the player. Darkness, a minor hazard, light rain, a nervous but untrained opponent. Should feel like a bump, not a wall.

2 DD

Genuinely difficult

Two stacked factors, or one serious one. A skilled opponent, a significant wound, working against the clock in bad conditions. This is the GM's standard "hard" setting.

3 DD

Dire

The situation is stacked against the player. Multiple serious obstacles in combination. Should feel desperate. Push back if the fiction doesn't clearly justify it.

4+ DD

Near-hopeless

Reserved for extreme situations: suicidal odds, legendary opposition, the worst possible moment to attempt something. Even 4 AD only wins 65.5% here (vs 4 DD).

QUICK ODDS LOOKUP — "WHAT ARE MY PLAYER'S CHANCES?"

Player's AD pool	0 DD	1 DD	2 DD	3 DD	4 DD	5 DD	6 DD
1 AD (weak / unskilled)	50%	42%	35%	29%	24%	20%	17%
2 AD (average)	75%	67%	59%	51%	44%	38%	33%
3 AD (capable / attribute fits)	88%	81%	74%	67%	60%	53%	47%
4 AD (strong advantage)	94%	90%	84%	78%	72%	66%	59%
5 AD (stacked advantages)	97%	94%	91%	86%	81%	75%	69%

Green comfortable Yes majority Amber genuine uncertainty Red failure is the likely outcome

DD SOURCES — BUILD FROM FACTORS, NOT FROM GUT FEEL

Environment

Darkness, difficult terrain, bad weather, time pressure, crowd, unstable surface — 1 DD per significant environmental factor

Opposition tags

Each relevant tag on the NPC or threat adds 1 DD. "Heavily armoured" adds 1 DD to attack. "Master swordsman" adds 1 DD to contests.

Character conditions

Each active condition relevant to the action adds 1 DD. Injured, exhausted, frightened, outnumbered, disarmed.

Situational gaps

Acting without proper tools, improvising, attempting outside competence, under direct fire. 1 DD per meaningful gap from ideal conditions.